

Site Selection & Patient Matching

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 04 of 8 · rank sites on real performance and size cohorts on RWD — humans select sites and confirm eligibility

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM SITE GUESSWORK TO EVIDENCE

Site Selection & Patient Matching

A governed feasibility copilot that ranks sites on historical performance and sizes cohorts against real-world data — while humans select the sites and clinicians confirm every patient's eligibility, with diversity surfaced as a first-class metric.

~80%

of trials miss their original enrollment timeline

[sector-press/estimate — flag on slide]

~20% / ~30%

of sites enroll zero / under-enroll patients

[sector-press/estimate]

~36%

average screen-failure rate (70–85% in CNS/rare disease)

[sector-press/estimate]

The Issue & The Cost of Doing Nothing

THE ISSUE

- Feasibility still leans on site-opinion surveys rather than historical performance data.
- Real-world cohort sizing is rarely integrated at scale, so enrollment assumptions are optimistic.
- ~1 in 5 sites enroll no one and ~30% under-enroll — wasted activation cost and lost time.
- FDA Diversity Action Plan expectations make representativeness a design requirement, not an afterthought.

THE COST OF DOING NOTHING

~\$24M / launch

Modeled: compressing enrollment by 30 days $\approx 30 \times \$800\text{K/day}$ lost sales $\approx$ ~\$24M recovered per launch. Cost per patient recruited commonly \$1,500–\$10,000; recruitment is 20–30% of trial budget.

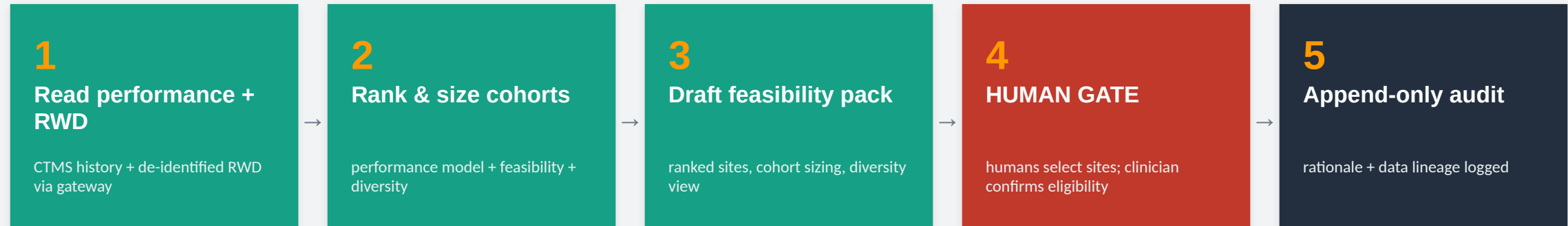
- Site activation spend on non-enrolling sites.
- Diversity-plan and ICH E17 gaps that jeopardize acceptability of the evidence.

[modeled — Tufts 2024 day-of-delay; recruitment benchmarks]

The design line: the agent ranks sites and sizes cohorts; humans select the sites and a clinician confirms each patient's eligibility — no autonomous enrollment.

How We Solve It — A Governed Pipeline

Read historical site performance and de-identified RWD through a scoped gateway, rank and size — then route selection and eligibility to humans.



Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

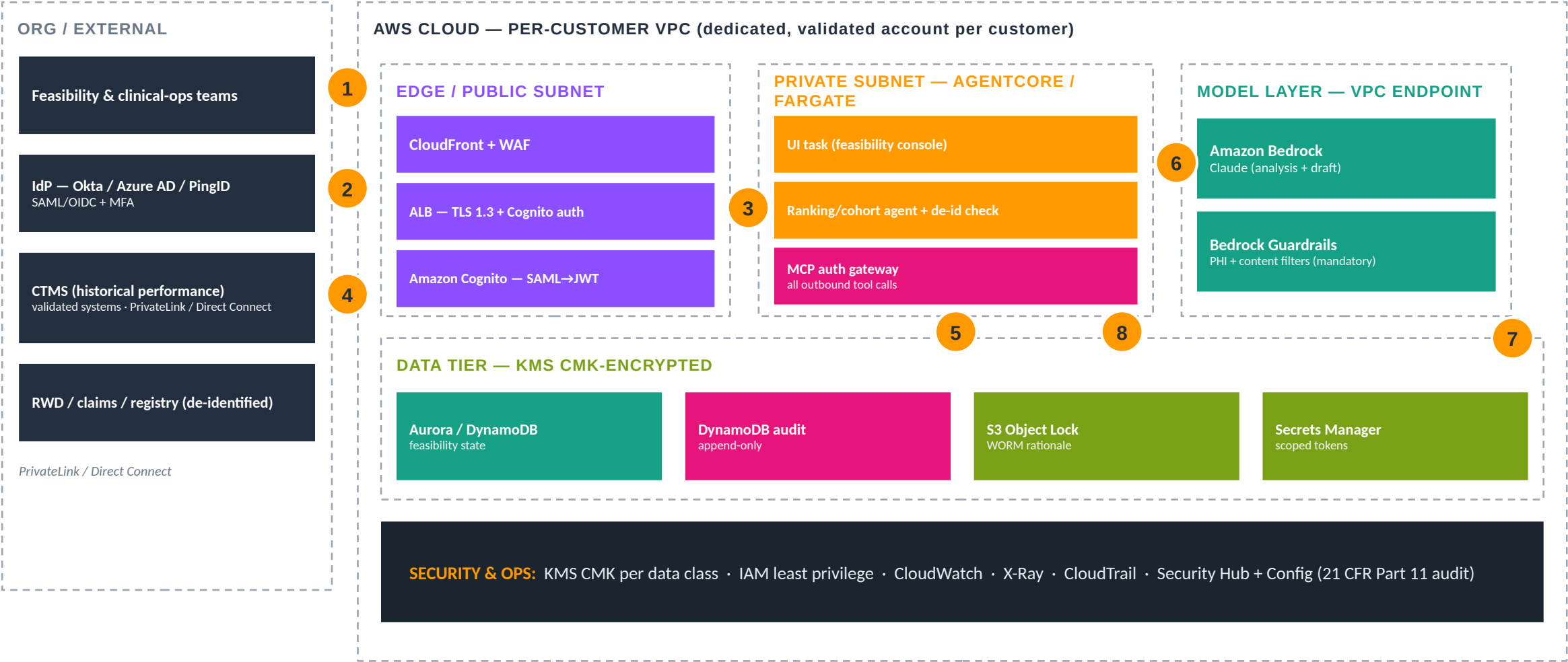
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides site selection & patient eligibility

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 user sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to feasibility console 4 CTMS + de-identified RWD reads over private connectivity 5 demand scales the agent workers 6 model calls stay in-VPC via Bedrock + Guardrails 7 ranking rationale + data lineage persisted to append-only audit 8 connectors reachable only through the governed MCP gateway (de-identification enforced)

Proof, Payback & How to Deploy

MEASURED OUTCOMES

~\$24M

modeled recovery per launch from a 30-day pull-in

[modeled]

diversity

representativeness surfaced as a first-class metric

[FDA Diversity Action Plan posture]

evidence

historical performance vs. opinion surveys

[design property]

de-identified

RWD via Safe Harbor / Expert Determination

[HIPAA]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (feasibility / clinician roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect CTMS history + de-identified RWD; enforce de-identification at the gateway.
1. Enable S3 Object Lock + append-only audit; run smoke + HITL test.

Deploy: CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway; point connectors at CTMS history and a de-identified RWD source.

[aws-native-reference/04-site-patient-matching/DEPLOY.md](#) · [docs/DEPLOY-QUICKSTART.md](#)

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.